DSC5001 STATISTICAL MACHINE LEARNING I

PROJECT PROPOSAL

From Forecast Accuracy to Inventory Cost: Evaluating Hierarchical Demand Forecasts on M5

Machine Not Learning | 27 October 2026

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RESEARCH SNAPSHOT

Decision problem. Which demand model and forecast level deliver the lowest inventory cost at a required service level? **Comparison.** Machine-learning gains are tested with and without hierarchical reconciliation. **Evidence.** Rolling-origin forecasts from M5 are translated into periodic-review inventory decisions and evaluated by cost, fill rate, and inventory, alongside accuracy metrics.

1. Problem and Research Question

Retailers forecast demand at several levels. Replenishment happens at the item-store level, while category and regional plans use aggregate forecasts. Reconciliation makes these levels coherent, but its effect on inventory cost has rarely been compared with gains from machine learning. The project asks two questions. First, do SKU-level machine-learning forecasts reduce inventory cost relative to statistical baselines? Second, does hierarchical reconciliation change that result? We also test whether the benefit depends on demand volume, intermittency, and the stockout-to-holding cost ratio.

The decision setting is a periodic-review order-up-to policy with a 28-day review period. For each series, the order target is set from a demand quantile. A planner needs the model and forecast level that produce the lowest cost at a required service level; lower error matters only if it supports that choice. The project is new and independent of other coursework or external projects.

2. Data

We use the M5 Walmart dataset, which contains 3,049 products sold in 10 stores across three US states. The item-store level has 30,490 daily series and 1,941 observations from 29 January 2011 to 19 June 2016. The data include 42,840 series across 12 aggregation levels, a calendar of holidays and events, and weekly selling prices (Makridakis et al., 2020, 2022). Access requires joining the competition on Kaggle and accepting its rules.

We will use 5,000 item-store series from California (CA_1-CA_4) for development and the full 30,490 series for the final run if compute allows. The 28-day horizon matches the competition evaluation window and a monthly replenishment cycle. The development sample will be selected before model estimation, and the final run will preserve the same hierarchy and forecast origins.

3. Method

For baselines, we use seasonal naive, exponential smoothing (ETS), and Croston or TSB for intermittent series. These provide a reference point for planners without machine learning. All methods will be fitted on identical forecast origins and series sets.

For the machine-learning models, we use LightGBM with lagged sales, rolling means and standard deviations, calendar features, price, and promotion indicators. We train point models and quantile models at $\tau = 0.5, 0.9, 0.95$, and 0.99. Features at each origin use only information available before that origin. To support top-down comparison, the same model family will also generate forecasts at the preselected aggregate levels.

To test reconciliation, we compare unreconciled forecasts with bottom-up, top-down, and MinT reconciliation (Wickramasuriya et al., 2019). This shows whether coherent forecasts improve decisions beyond their effect on aggregate error. We will report coherence checks on the aggregation identity and preserve quantile ordering after reconciliation.

For the inventory simulation, we translate forecasts into orders under a periodic-review order-up-to policy. The critical fractile is $q^* = Cu / (Cu + Co)$, where Cu is the stockout cost and Co is the holding cost. We scan service levels from 0.80 to 0.99 and multiple cost ratios, and report the resulting cost curve rather than relying on a single assumed ratio.

Validation uses rolling-origin evaluation with several 28-day windows. We report WAPE, MASE, and pinball loss for forecasts, and total inventory cost, fill rate, and average inventory for decisions. Inventory-cost differences will be accompanied by uncertainty estimates from a fixed, reproducible resampling procedure.

4. Expected Contribution

The project will produce a cost-based comparison of model classes and forecast hierarchies on the same data. Its main output is a decision map: for a given service level and demand pattern, which forecast level and model produce the lowest inventory cost. We expect accuracy gains to be largest for high-volume, low-intermittency series; the experiment will show whether those gains survive the cost calculation (Makridakis et al., 2022; Spiliotis et al., 2021).

The result should help a retail planner choose between a simple aggregate forecast and a more expensive SKU-level model. A null cost result would also be informative if it showed where higher forecast accuracy does not materially improve the inventory decision.

5. Work Plan and Team Division

The project is organized into five linked work packages. Xiaofeng ZHU leads data preparation and leakage-safe feature construction; Jiaheng LI leads baselines and LightGBM point and quantile models; Junze LU leads hierarchy construction and bottom-up, top-down, and MinT reconciliation; Xiang LI leads rolling-origin evaluation and inventory-cost simulation; Jianfeng LIN leads the report, slides, recording, and cross-artifact consistency. Each package has a named lead, a versioned output, and an agreed acceptance check. All members contribute to code review, interpretation, and the presentation.

Weeks 1 and 2 cover the data audit, feature pipeline, baselines, and first model runs. Weeks 2 and 3 cover LightGBM point and quantile forecasts, hierarchy reconciliation, and the first inventory simulation. Week 4 covers the full rolling-origin run, sensitivity analysis, decision-map figures, report draft, slides, and recording. Final QA checks that the proposal, code, results, appendix, and presentation use the same sample and metric definitions.

All code will be version-controlled in a shared GitHub repository from Week 1; the repository link will be included in the final Appendix.

6. Risks and Mitigation

M5 is a public benchmark with many published results. The contribution therefore depends on the cost and hierarchy framing, and we will report the full cost curve rather than a single accuracy score. Inventory costs are assumptions, so we will test a range of Cu and Co values. The full dataset is large; we will develop on a fixed subset and document the final run. Forecast reconciliation adds implementation complexity, so coherence and quantile constraints will be checked before the forecasts enter the inventory simulation.

7. Success Criteria

We will treat the project as successful if cost curves differ by more than the stated simulation or resampling uncertainty, model rankings remain reasonably stable across rolling windows and sensitivity settings, and the decision map identifies at least one demand segment where a simpler model or forecast level matches the machine-learning approach. A null result on cost, alongside a stable accuracy gain, would still be useful: it would show where forecast accuracy does not reach the inventory decision.

References

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). *M5 Forecasting - Accuracy* [Data set]. Kaggle. https://www.kaggle.com/competitions/m5-forecasting-accuracy

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). The M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting, 38*(4), 1346-1364. https://doi.org/10.1016/j.ijforecast.2021.11.013

Spiliotis, E., Makridakis, S., Kaltsounis, A., & Assimakopoulos, V. (2021). Product sales probabilistic forecasting: An empirical evaluation using the M5 competition data. *International Journal of Production Economics, 240*, Article 108237. https://doi.org/10.1016/j.ijpe.2021.108237

Wickramasuriya, S. L., Athanasopoulos, G., & Hyndman, R. J. (2019). Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization. *Journal of the American Statistical Association, 114*(526), 804-819. https://doi.org/10.1080/01621459.2018.1448825